

Lab Report 2 - Cybersecurity Analytics

Performing a live capture in Wireshark

Introduction

This lab is designed to work with VM machines to get practical skills in network traffic capture, filtering, and analysis using Wireshark. Wireshark can capture traffic from many different network media types, including Ethernet, Wireless LAN, Bluetooth, USB, and more (Wireshark Foundation, n.d.). Packet analysis provides important evidence for digital forensics, and Wireshark allowing analytics to restructure communication flows and verify system behavior. Packet analysis describes the process of capturing and interpreting live data as it flows across a network in order to better understand what is happening on that network (Sanders, 2017).

In this lab, the Kali Linux OS is used to capture live packets using Wireshark on the eth0 interface while browsing several websites including Google.com, MSUM website, Northland college website, and Amazon website. Captured traffic from these websites about 5 minutes and save it as .pcap file to observe and filter the packets. After filtering the packets, confirm the details of IP addresses using *whois* website. This lab helps to understand how network activities can be monitored, how to trace communication flows and get the endpoint details and validate them using known web services.

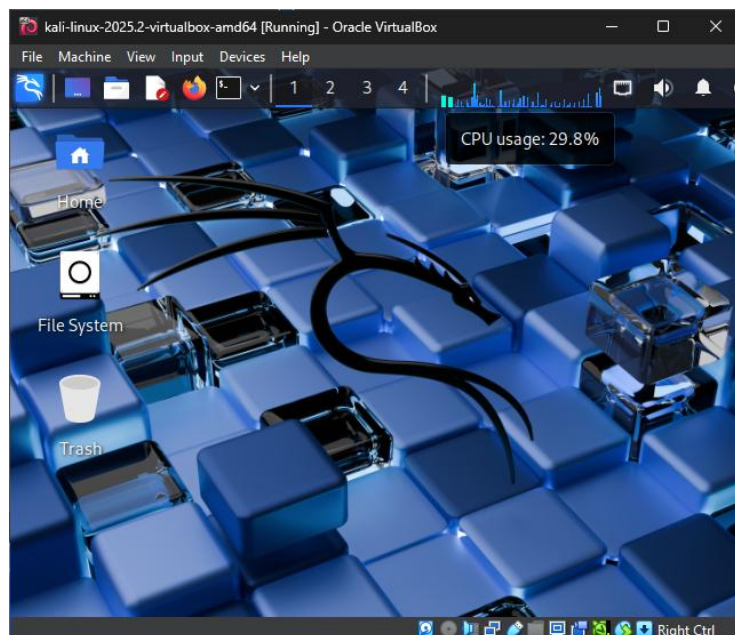
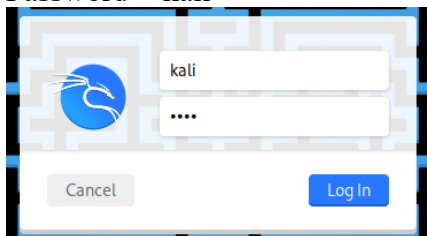
Step - 1

1. Logged to Azure Lab environment using following web link and logged into the Kali VM using Oracle VirtualBox.

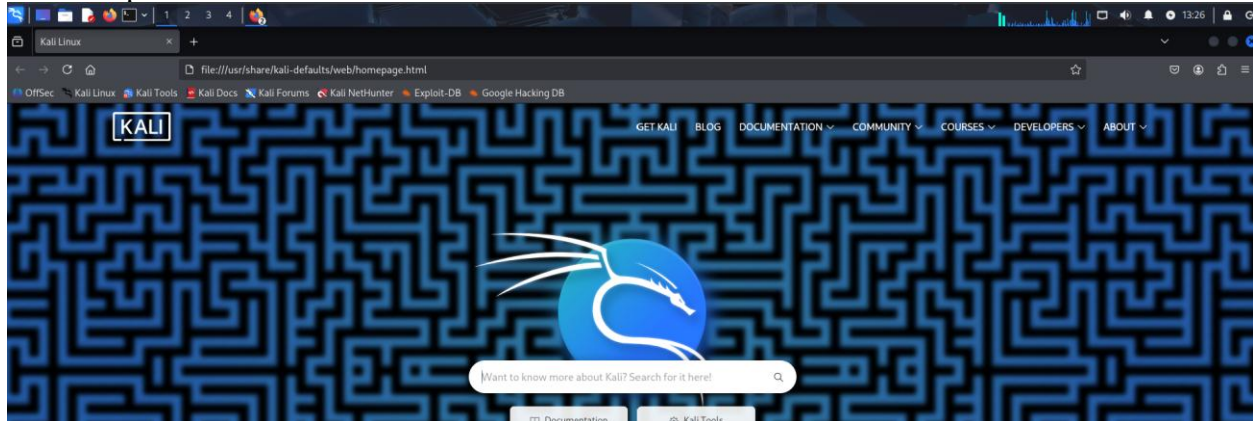
https://labs.azure.com/virtualmachines?feature_vnext=true

Username - kali

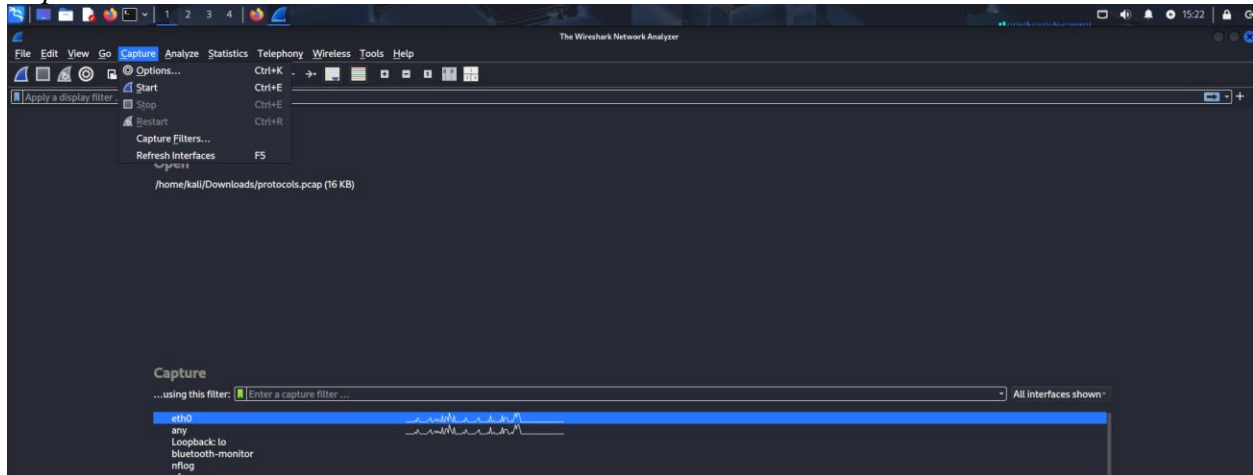
Password - kali



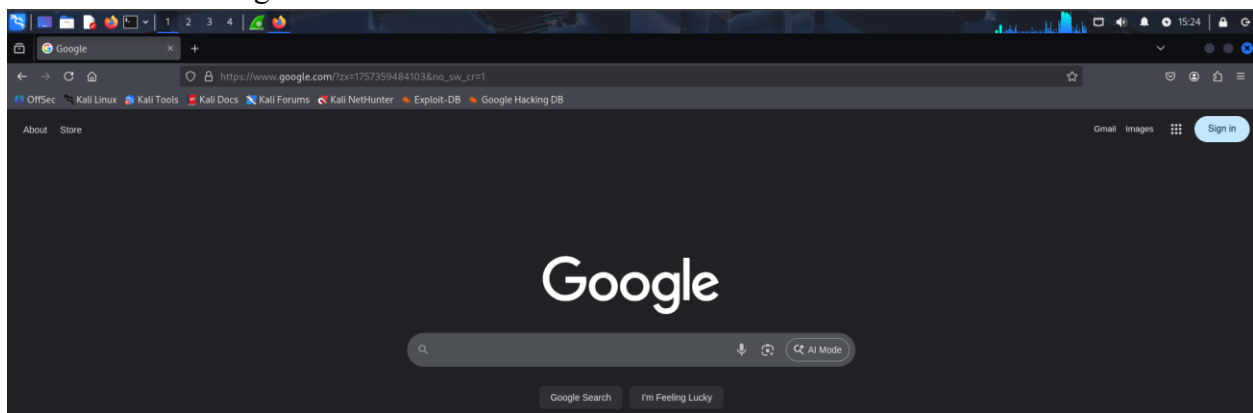
2. Opened the Firefox web browser.



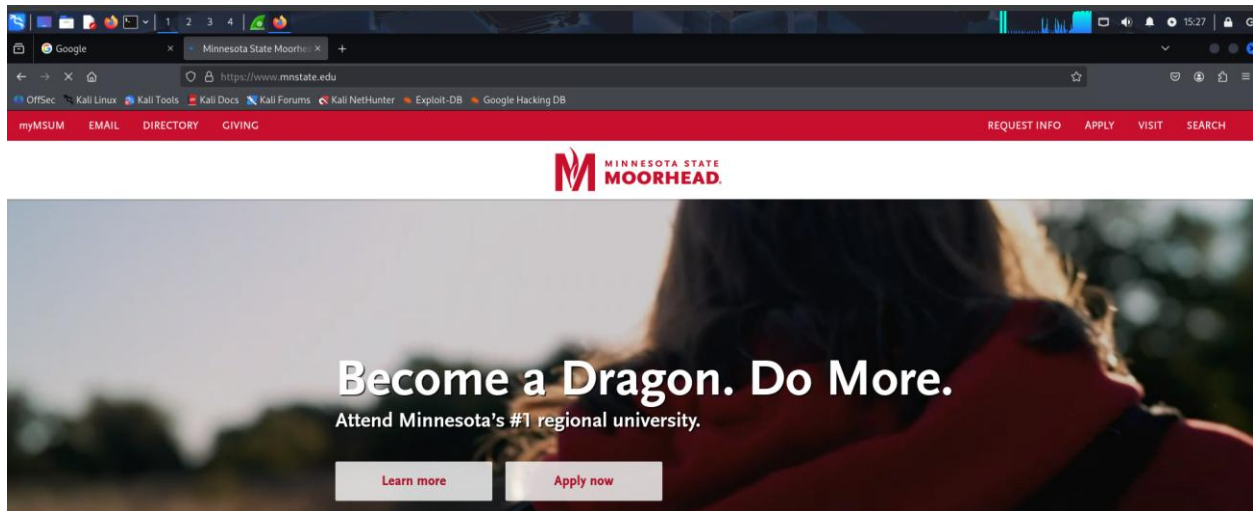
3. Opened Wireshark and selected *eth0* interface and started a live capture by clicking on *Capture* menu and then clicked *start*.



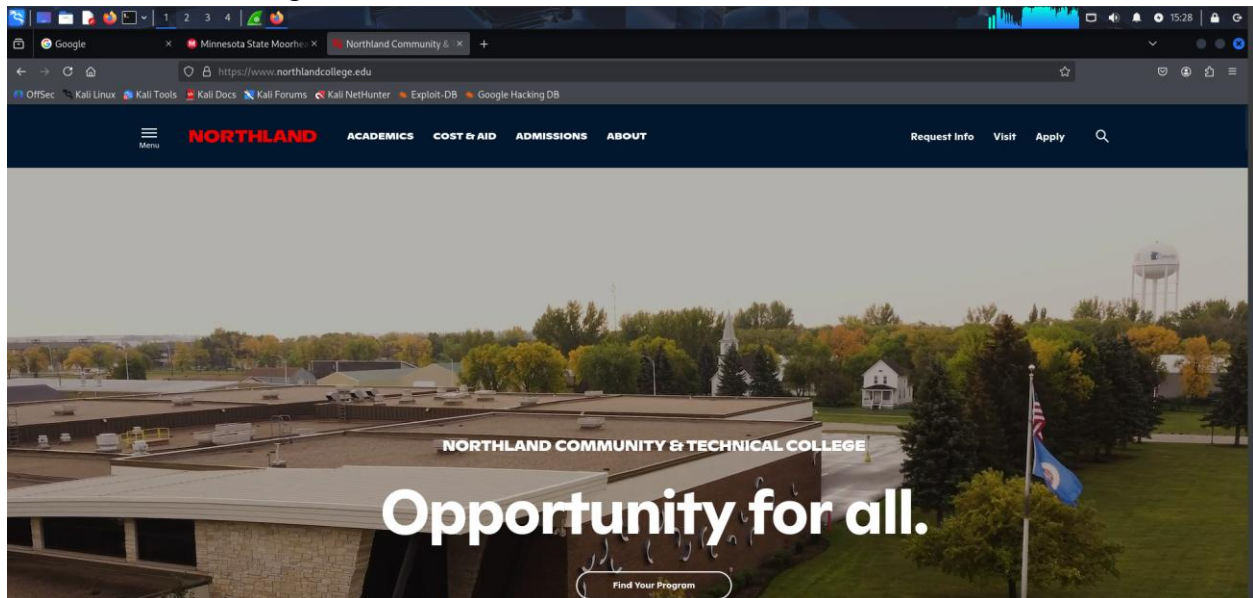
4. Opened Firefox and visited to
a. Google.com



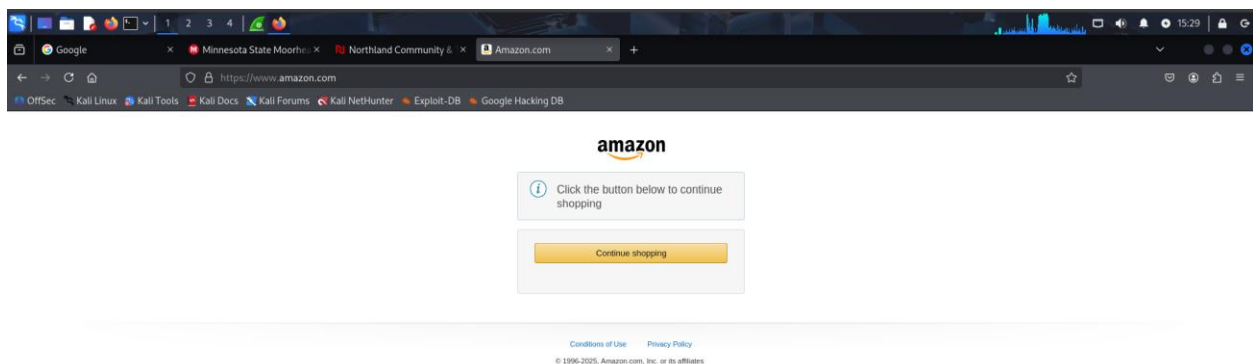
b. mnstate.edu



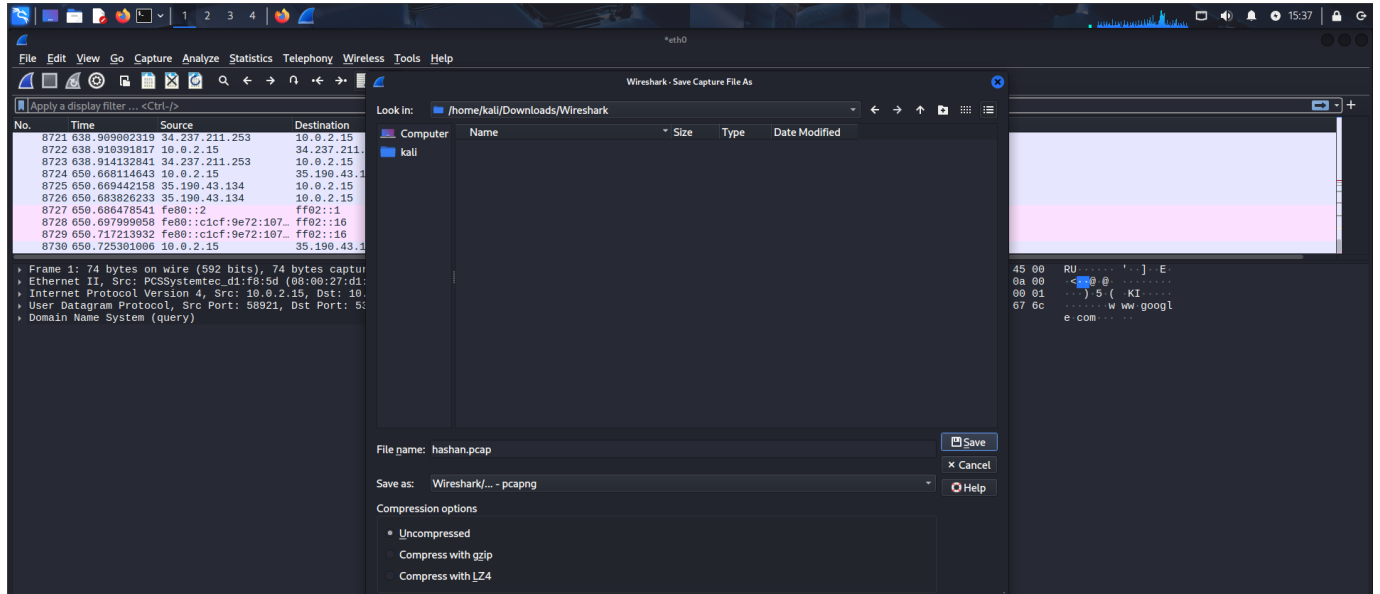
c. northlandcollege.edu



d. amazon.com

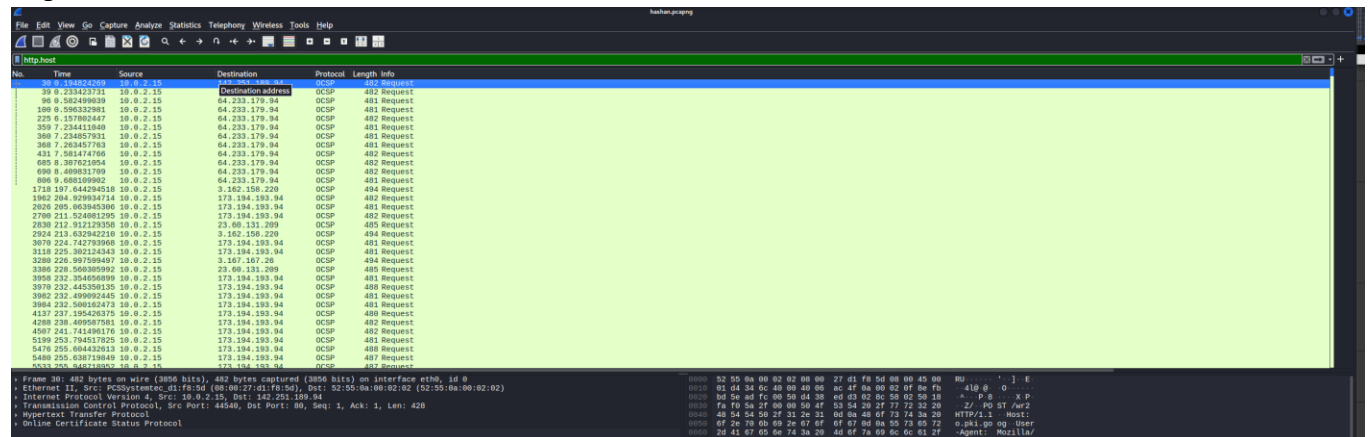


5. Stopped the capture and save the file as *hashan.pcap* in */home/kali/Downloads/Wireshark* folder.

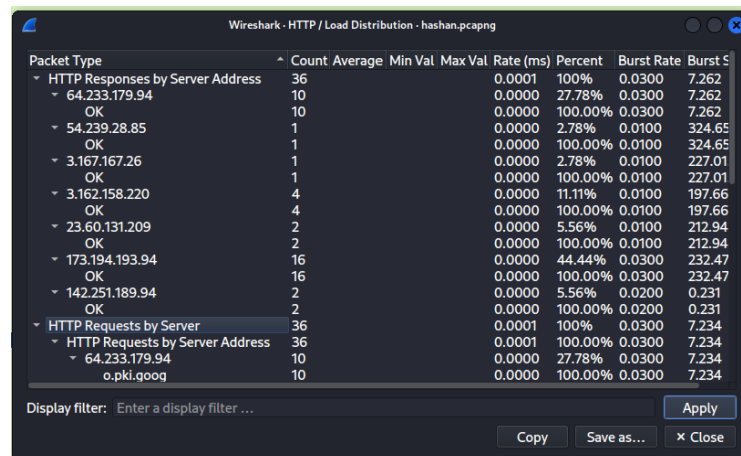


Step -2

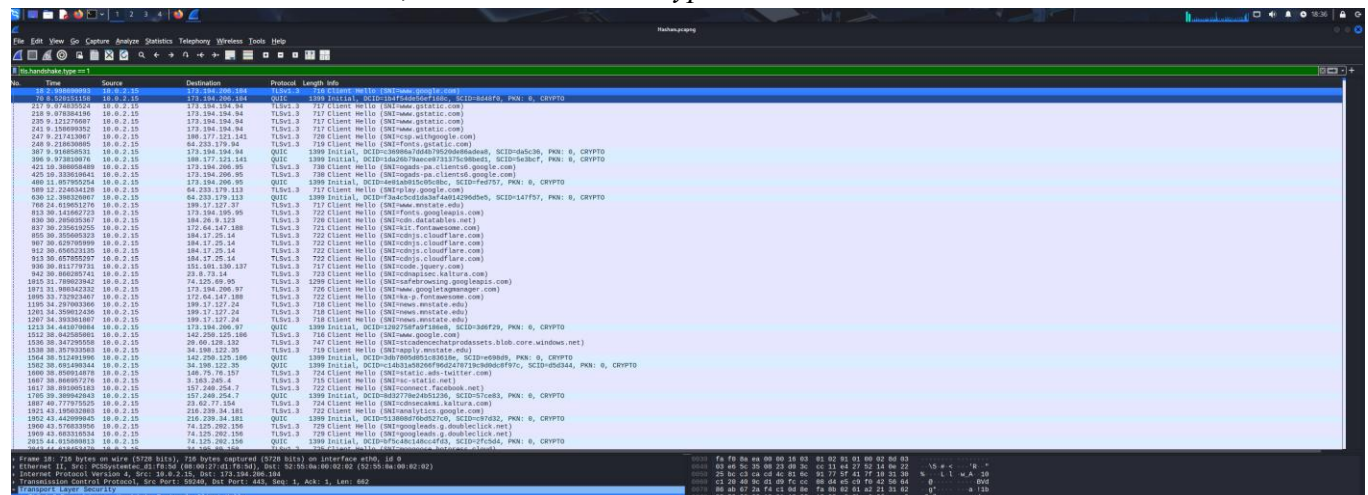
1. Open the file in Wireshark and enter *http.host* in the filter and apply the filter to isolate the http host details.



2. Clicked on the *Statistics* menu. Clicked on *Load Distribution* and then captured the result.



3. This network uses HTTPS protocol, So the HTTP filter has encrypted data. To find the details of website IP addresses, used *tls.handshake.type == 1* filter.



- The screenshot shows a Wireshark interface with a packet capture of DNS traffic. The top pane displays a list of packets, with packet 10 selected. The details pane shows the structure of the selected packet, which is a Standard query response from 193.174.206.103 to 193.174.206.104. The packet bytes pane shows the raw data of the packet.

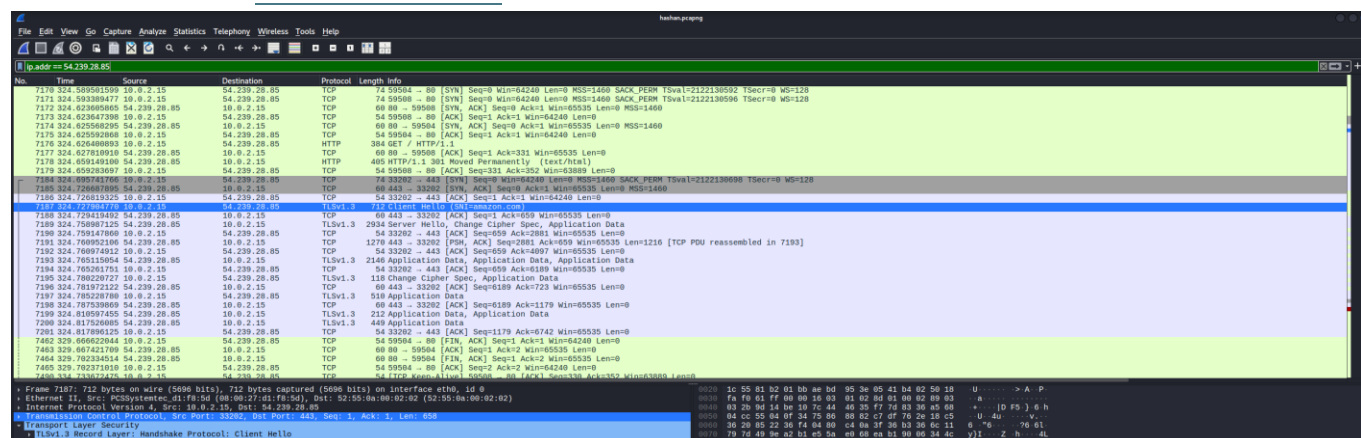
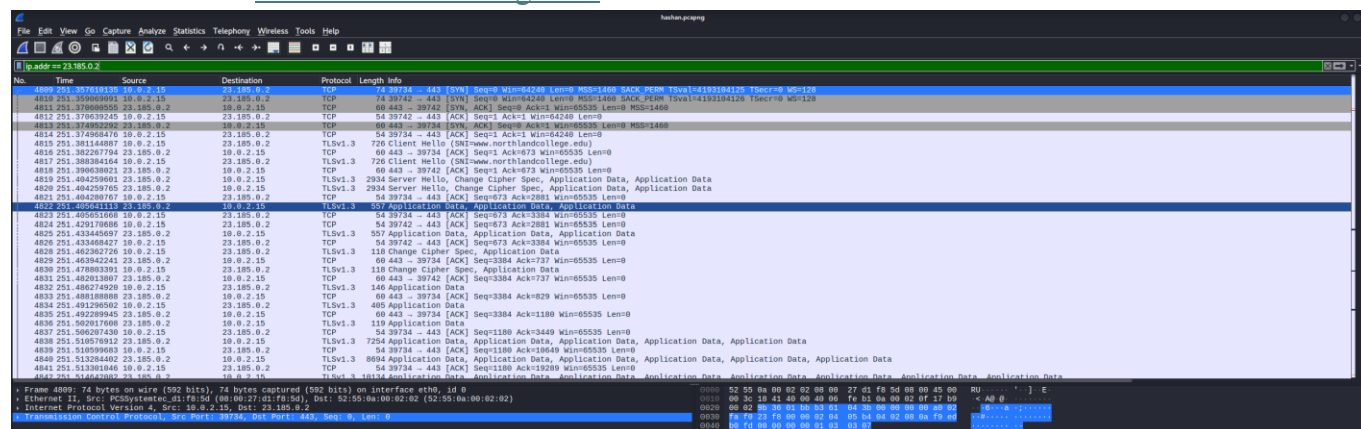
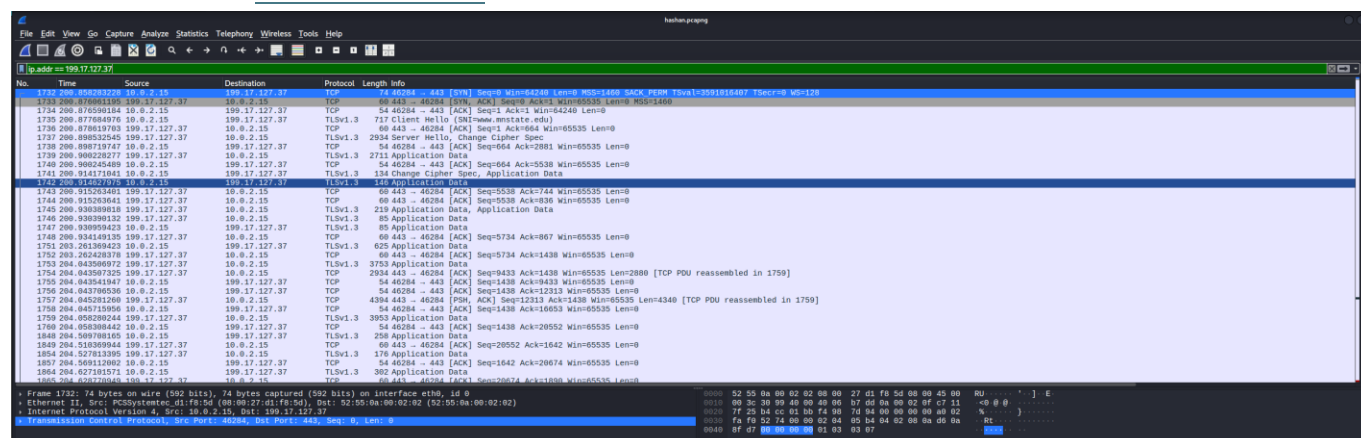
-
- The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for file operations, capture control, and analysis. The main display area is divided into three panes:
- Packet List Pane:** Shows a list of captured packets. The selected packet is No. 1730, which is a DNS Standard query response from 10.0.2.15 to 192.168.2.10.
 - Packet Details Pane:** Displays the hierarchical structure of the selected packet. It shows the Ethernet II frame, Internet Protocol Version 4, and the Hypertext Transfer Protocol.
 - Packet Bytes Pane:** Shows the raw data of the selected packet in hexadecimal and ASCII format.
- The packet list pane contains the following information:
- | No. | Time | Source | Destination | Protocol | Length | Info |
|------|-------------|-----------|-------------|----------|--------|--|
| 1729 | 2.966430554 | 10.0.2.15 | 10.0.2.10 | DNS | 75 | Standard query 0x7cd1 AAAA www.msstate.edu |
| 1730 | 2.968415153 | 10.0.2.15 | 10.0.2.10 | DNS | 75 | Standard query 0x7cd1 AAAA www.msstate.edu |
| 1731 | 2.969797113 | 10.0.2.3 | 10.0.2.15 | DNS | 138 | Standard query response 0x7cd1 AAAA www.msstate.edu SOA ns1-01.azure-dns.com |
| 1732 | 2.970973386 | 10.0.2.3 | 10.0.2.15 | DNS | 91 | Standard query response 0x8dc AAAA www.msstate.edu A 199.17.127.37 |

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- The screenshot shows the Wireshark interface with a packet list and packet details pane. The packet list shows four DNS-related packets:
- | No. | Time | Source | Destination | Protocol | Length | Info |
|------|---------------|-----------|-------------|----------------|--------|---|
| 4806 | 251.358878749 | 10.0.2.15 | 10.0.2.3 | Standard query | 82316 | A www.northlandcollege.edu |
| 4806 | 251.359133149 | 10.0.2.15 | 10.0.2.3 | DNS | 84 | Standard query 8x8714 AAAA www.northlandcollege.edu |
| 4807 | 251.361643808 | 10.0.2.3 | 10.0.2.15 | DNS | 180 | Standard query response 8x2316 A www.northlandcollege.edu A 23.105.0.2 |
| 4808 | 251.368878749 | 10.0.2.15 | 10.0.2.3 | DNS | 1481 | Standard query response 8x8714 AAAA www.northlandcollege.edu AAAA 2020:12:8090:12:AAAA 2020:12:8090:12:AAAA |
- The packet details pane for the selected packet (No. 4806) shows the following structure:
- Standard query query 8x2316 A www.northlandcollege.edu
 - Transaction ID: 0
 - Flags: 0x0000 (Standard query query)
 - Questions: 1
 - Question 1: www.northlandcollege.edu IN A
 - Answers: 0
 - Authorities: 0
 - Additional: 0

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- The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for file operations, navigation, and analysis. The main display area is divided into three panes:
- Packet List Pane:** Shows a list of captured packets. The first packet is a DNS Standard query from 192.168.1.105 to 10.0.2.15. Subsequent packets show a series of DNS queries and responses between the same IP addresses.
 - Packet Details Pane:** Displays the hierarchical structure of the selected packet (DNS Standard query response). It shows fields like Flags, Transaction ID, and Question section.
 - Packet Bytes Pane:** Shows the raw data of the selected packet in hexadecimal and ASCII format.
- The packet list pane contains the following entries:
- | No. | Time | Source | Destination | Protocol | Length | Info |
|------|----------------|-----------|-------------|----------|--------|---|
| 7263 | 7263.654215129 | 10.0.2.15 | 10.0.2.15 | DNS | 74 | Standard query 6a524a to www.google.com |
| 7263 | 7263.654595566 | 10.0.2.15 | 10.0.2.3 | DNS | 74 | Standard query 6a5338 AAAA www.amazon.com |
| 7264 | 7264.656233116 | 10.0.2.3 | 10.0.2.15 | DNS | 169 | Standard query response 6a524a to www.google.com CNAME pt.47c72cb9-frontier.amazon.com CNAME d3agbhuk4h2yn.cloudfront.net A 18.154.117.229 |
| 7265 | 7265.937984960 | 10.0.2.3 | 10.0.2.15 | DNS | 377 | Standard query response 6a5338 AAAA www.amazon.com CNAME pt.47c72cb9-frontier.amazon.com CNAME d3agbhuk4h2yn.cloudfront.net AAAA 2000:2000:2000:2000:749a:5f6d:9081 |

- The screenshot displays the Wireshark network protocol analyzer interface. The top pane (Packet List) shows a list of captured packets, with packet 64 selected. The middle pane (Packet Details) shows the structure of the selected packet, which is a TLS Handshake (Seq=643). The bottom pane (Packet Bytes) shows the raw data of the packet in hexadecimal and ASCII format.

No.	Time	Source	Destination	Protocol	Length	Info
64	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
65	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
66	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
67	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
68	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
69	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
70	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
71	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
72	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
73	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
74	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
75	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
76	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
77	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
78	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
79	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
80	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
81	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
82	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
83	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
84	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
85	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
86	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
87	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
88	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
89	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
90	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
91	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
92	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
93	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
94	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
95	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
96	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
97	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
98	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
99	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data
100	0.000000	172.194.206.99	172.194.206.99	TLSv1.3	1275	Application Data



Source and destination IP address details

Website	Source IP	Destination IP	Protocols	Ports Used
www.google.com	10.0.2.15	173.194.206.104	TLS/HTTPS, TCP, DNS	43900, 443
www.mnstate.edu	10.0.2.15	199.17.127.37	TLS/HTTPS, TCP, DNS	46284, 443
www.northlandcollege.edu	10.0.2.15	23.185.0.2	TLS/HTTPS, TCP, DNS	39734, 443
www.amazon.com	10.0.2.15	54.239.28.85	TLS/HTTPS, TCP, DNS	33202, 443

DNS (Domain Name System) - Used to resolve website names (e.g., google.com) into their IP addresses. This is why the first few packets are DNS queries to external DNS servers

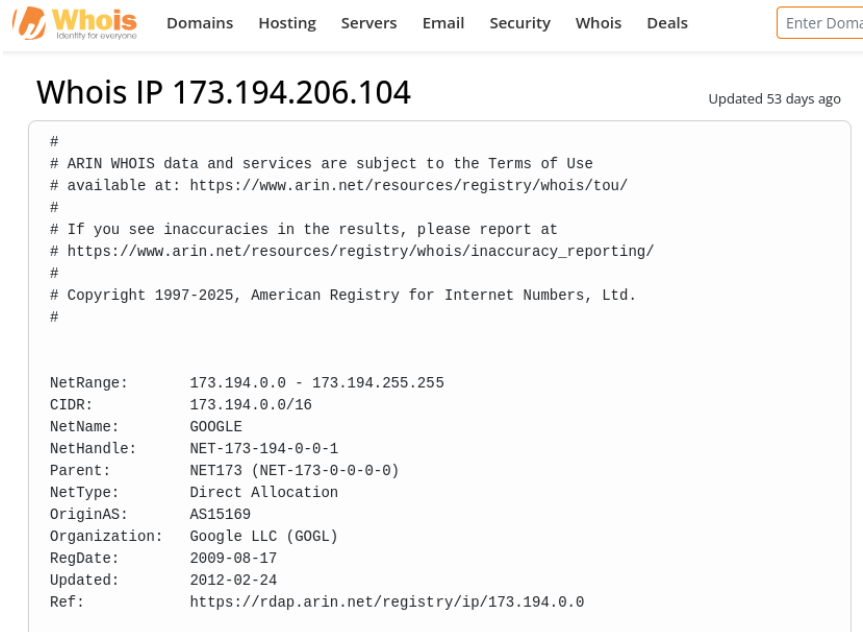
TCP (Transmission Control Protocol) - Provides reliable connections between the Kali VM and the destination servers. Almost all web traffic runs over TCP.

HTTPS (TLS/SSL) – Most of the browsing on nowadays use HTTPS. Wireshark shows these as *TLSv1.2* or *TLSv1.3* sessions, protecting communication confidentiality by encrypting data.

Step 3 -

1. Checked all the IP addresses with <https://www.whois.com/whois/> and take capture the results.

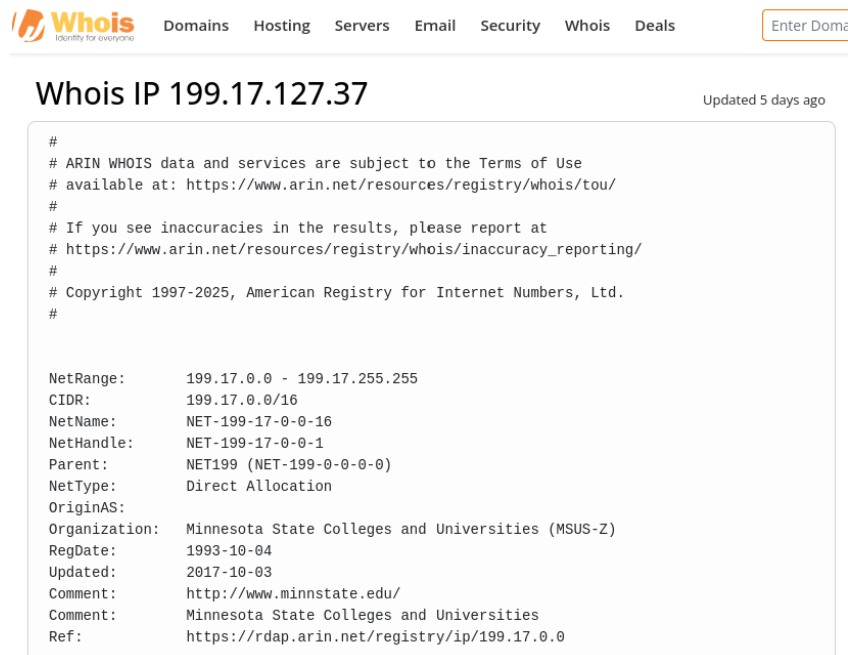
a. For IP address 173.194.206.104



The screenshot shows the Whois website interface. At the top, there is a navigation bar with links: Domains, Hosting, Servers, Email, Security, Whois, and Deals. A search bar on the right contains the text "Enter Domain". Below the navigation bar, the title "Whois IP 173.194.206.104" is displayed, followed by "Updated 53 days ago". The main content area contains a block of text starting with "# ARIN WHOIS data and services are subject to the Terms of Use" and ending with "# Copyright 1997-2025, American Registry for Internet Numbers, Ltd.". Below this, the following information is listed:

```
NetRange: 173.194.0.0 - 173.194.255.255
CIDR: 173.194.0.0/16
NetName: GOOGLE
NetHandle: NET-173-194-0-0-1
Parent: NET173 (NET-173-0-0-0-0)
NetType: Direct Allocation
OriginAS: AS15169
Organization: Google LLC (GOGL)
RegDate: 2009-08-17
Updated: 2012-02-24
Ref: https://rdap.arin.net/registry/ip/173.194.0.0
```

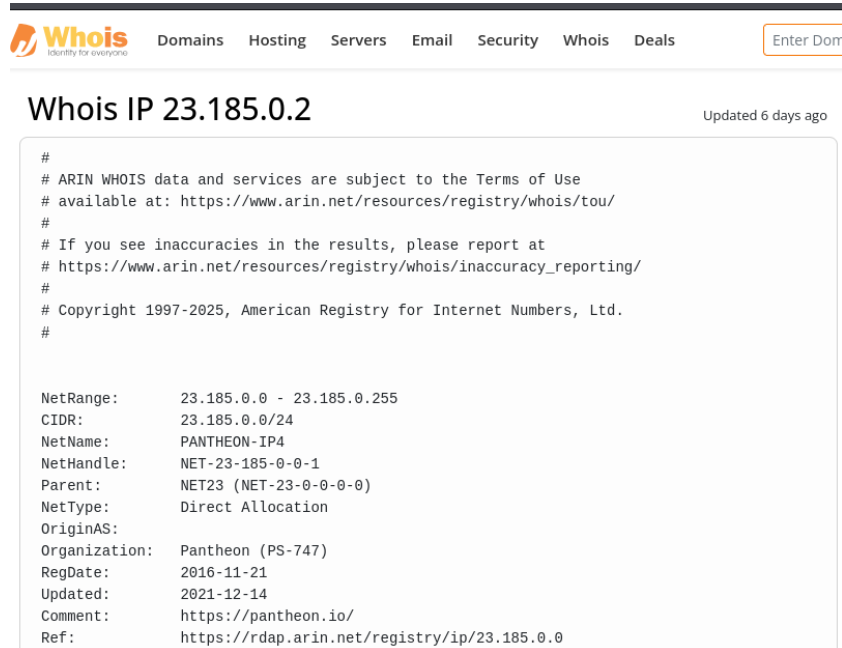
b. For IP address 199.17.127.37



The screenshot shows the Whois website interface. At the top, there is a navigation bar with links: Domains, Hosting, Servers, Email, Security, Whois, and Deals. A search bar on the right contains the text "Enter Domain". Below the navigation bar, the title "Whois IP 199.17.127.37" is displayed, followed by "Updated 5 days ago". The main content area contains a block of text starting with "# ARIN WHOIS data and services are subject to the Terms of Use" and ending with "# Copyright 1997-2025, American Registry for Internet Numbers, Ltd.". Below this, the following information is listed:

```
NetRange: 199.17.0.0 - 199.17.255.255
CIDR: 199.17.0.0/16
NetName: NET-199-17-0-0-16
NetHandle: NET-199-17-0-0-1
Parent: NET199 (NET-199-0-0-0-0)
NetType: Direct Allocation
OriginAS:
Organization: Minnesota State Colleges and Universities (MSUS-Z)
RegDate: 1993-10-04
Updated: 2017-10-03
Comment: http://www.minnstate.edu/
Comment: Minnesota State Colleges and Universities
Ref: https://rdap.arin.net/registry/ip/199.17.0.0
```

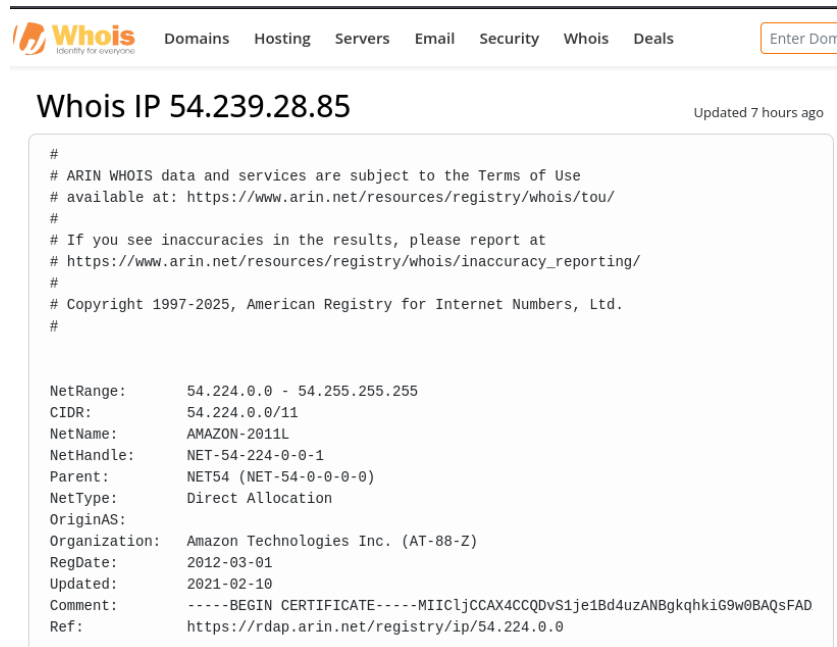
c. For IP address 23.185.0.2 - The DNS resolution for *www.northlandcollege.edu*, given the IP address 23.185.0.2. A whois.com website shows that this address is registered to Pantheon Systems, which indicates that Northland College hosts its website through Pantheon's managed infrastructure.



The screenshot shows the Whois.com website interface. The header includes the Whois logo and navigation links: Domains, Hosting, Servers, Email, Security, Whois, and Deals. A search bar on the right contains the text "Enter Dom". The main content area is titled "Whois IP 23.185.0.2" and indicates it was "Updated 6 days ago". Below the title, there is a block of text containing ARIN WHOIS data and services subject to the Terms of Use, available at <https://www.arin.net/resources/registry/whois/tou/>. It also includes a link for reporting inaccuracies at https://www.arin.net/resources/registry/whois/inaccuracy_reporting/ and a copyright notice for 1997-2025, American Registry for Internet Numbers, Ltd. Below this, a table of IP details is displayed:

NetRange:	23.185.0.0 - 23.185.0.255
CIDR:	23.185.0.0/24
NetName:	PANTHEON-IP4
NetHandle:	NET-23-185-0-0-1
Parent:	NET23 (NET-23-0-0-0-0)
NetType:	Direct Allocation
OriginAS:	
Organization:	Pantheon (PS-747)
RegDate:	2016-11-21
Updated:	2021-12-14
Comment:	https://pantheon.io/
Ref:	https://rdap.arin.net/registry/ip/23.185.0.0

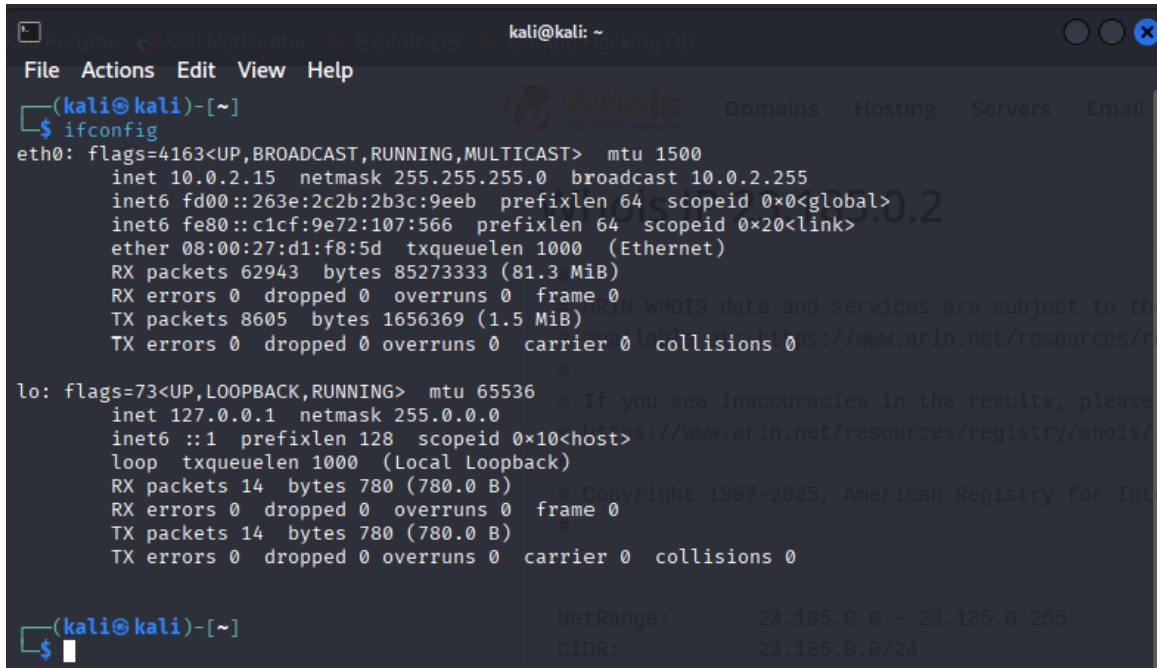
d. For IP address 54.239.28.85



The screenshot shows the Whois.com website interface. The header includes the Whois logo and navigation links: Domains, Hosting, Servers, Email, Security, Whois, and Deals. A search bar on the right contains the text "Enter Dom". The main content area is titled "Whois IP 54.239.28.85" and indicates it was "Updated 7 hours ago". Below the title, there is a block of text containing ARIN WHOIS data and services subject to the Terms of Use, available at <https://www.arin.net/resources/registry/whois/tou/>. It also includes a link for reporting inaccuracies at https://www.arin.net/resources/registry/whois/inaccuracy_reporting/ and a copyright notice for 1997-2025, American Registry for Internet Numbers, Ltd. Below this, a table of IP details is displayed:

NetRange:	54.224.0.0 - 54.255.255.255
CIDR:	54.224.0.0/11
NetName:	AMAZON-2011L
NetHandle:	NET-54-224-0-0-1
Parent:	NET54 (NET-54-0-0-0-0)
NetType:	Direct Allocation
OriginAS:	
Organization:	Amazon Technologies Inc. (AT-88-Z)
RegDate:	2012-03-01
Updated:	2021-02-10
Comment:	-----BEGIN CERTIFICATE-----MIICljCCAX4CCQDvS1je1Bd4uzANBgkqhkiG9w0BAQsFAD
Ref:	https://rdap.arin.net/registry/ip/54.224.0.0

2. IP address of your Kali Linux VM = 10.0.2.15
3. Using *ifconfig* command on command prompt we can find the IP address of Kali VM machine. Details as,
IP Address - 10.0.2.15
Network Mask – 255.255.255.0



```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd00::263e:2c2b:2b3c:9eeb prefixlen 64 scopeid 0x0<global>
    inet6 fe80::c1cf:9e72:107:566 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:d1:f8:5d txqueuelen 1000 (Ethernet)
    RX packets 62943 bytes 85273333 (81.3 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8605 bytes 1656369 (1.5 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 14 bytes 780 (780.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 14 bytes 780 (780.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

NetRange: 23.185.0.0 - 23.185.0.255
CIDR: 23.185.0.0/24
```

Conclusion

This lab helps to cover the practical aspects of the live packet analysis by using Wireshark. First, I visited four websites and captured live packets and saved them. By using several filters, I was able to filter out IP addresses of source and destinations, identify protocol and ports used for communication. These kinds of details are important for the security analyst to get the detail idea about the network and work around with troubleshooting.

Most of the browsing on nowadays use HTTPS. Wireshark shows these as TLSv1.2 or TLSv1.3 sessions, protecting communication confidentiality by encrypting data. DNS queries are useful to resolve domain names into the IP addresses. The sniffer verifies the protocol of the captured network data based on the information extracted and begins its analysis of that protocol's specific features (Sanders, 2017). Whois.com website is useful to get detail descriptions about any IP address you want to discover. For the Kali Linux, *ifconfig* command can be used to get the IP address details used by the VM. Packet level analysis provides greater support during troubleshooting network compromises and observes the performance of the system.

References

1. Sanders, C. (2017). *Practical packet analysis* (3rd ed.).
2. Wireshark Foundation. (n.d.). *Chapter 1. Introduction—What is Wireshark?* In *Wireshark user's guide*. Retrieved September 8, 2025, from https://www.wireshark.org/docs/wsug_html_chunked/ChapterIntroduction.html.
3. Minnesota State University Moorhead. (n.d.). *Minnesota State University Moorhead*. Retrieved September 8, 2025, from <https://www.mnstate.edu/>
4. Northland College. (n.d.). *Northland College*. Retrieved September 8, 2025, from <https://www.northlandcollege.edu/>
5. Whois.com. (n.d.). *Domain and IP lookup service*. Retrieved September 8, 2025, from <https://www.whois.com/>
6. Google. (n.d.). *Google*. Retrieved September 8, 2025, from <https://www.google.com/>
7. Amazon. (n.d.). *Amazon*. Retrieved September 8, 2025, from <https://www.amazon.com/>